KCCITM NOIDA • AKTU

WEEK 2 MODULE

PROGRAMMING PRACTICE

Name	:	Shruti Kumari Singh
Roll No.	:	2404920100331
Year	:	B.Tech 3rd Year
Branch	:	CSE

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Module 2

1) Fraud Transaction detection [Linear search]

<u>code</u>:

```
Public Class Fraud Transaction Detection {
  Public static void main (String [ ] args) {
    Scanner sc = new Scanner (system.in);
     int n = sc.next Int ();
   int [ ] transaction = new int [n];
   for (int i = 0; i<n; i++) {
        transaction [i] = sc. nextInt ();
   }
   int suspicious ID = sc. next Int ();
   boolean flag = false;
   for (int i : transaction) {
     if (i == suspicious ID) {
         Flag = true;
         break;
     }
   } if (flag) {
     system.out.println ("Fraud Transaction found ");
   } else {
    system.out.println (" Fraud Transaction Not found ");
   }
   sc. close ();
} }
```

```java
public class FraudTransactionDetection {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] transaction = new int[n];
        for (int i = 0; i < n; i++) {
            transaction[i] = sc.nextInt();
        }
        int suspiciousID = sc.nextInt();
        boolean Flag  = false;
        for (int i: transaction){
            if( i == suspiciousID){
                Flag = true;
                break;
            }
        }if(Flag){
            System.out.println("Fraud Transaction Found");
        }else{
            System.out.println("Fraud Transaction Not Found");
        }
        sc.close();
    }}
```

```
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found
```

2. Server log search [binary search]

```
Public class server log search {
 Public static void main ( String [] args ) {
    Scanner sc = new scanner ( system.in);
    int n = sc. nextInt ( );

  int [ ] logs = new int [n];
    for(int i = 0; i<n; i++) {
        logs[i] = sc. nextInt ();

    }
   int target = sc. next Int ( );
   boolean found = false;
   int start = 0;
   int end = n-1;
   while ( start <= end) {
      int mid = start +( end - start) / 2;
      if (logs [mid] == target ) {
          found = true;
          break;
      } else if ( logs [mid] < target ) {
          start = mid + 1;

      } else {
         end = mid - 1;

      }
  } if (found) {
         system. out. println (" log found ");
  } else {
      system. out. println (" log not found ");
```

collegeTraining Version control ServerLogSearch

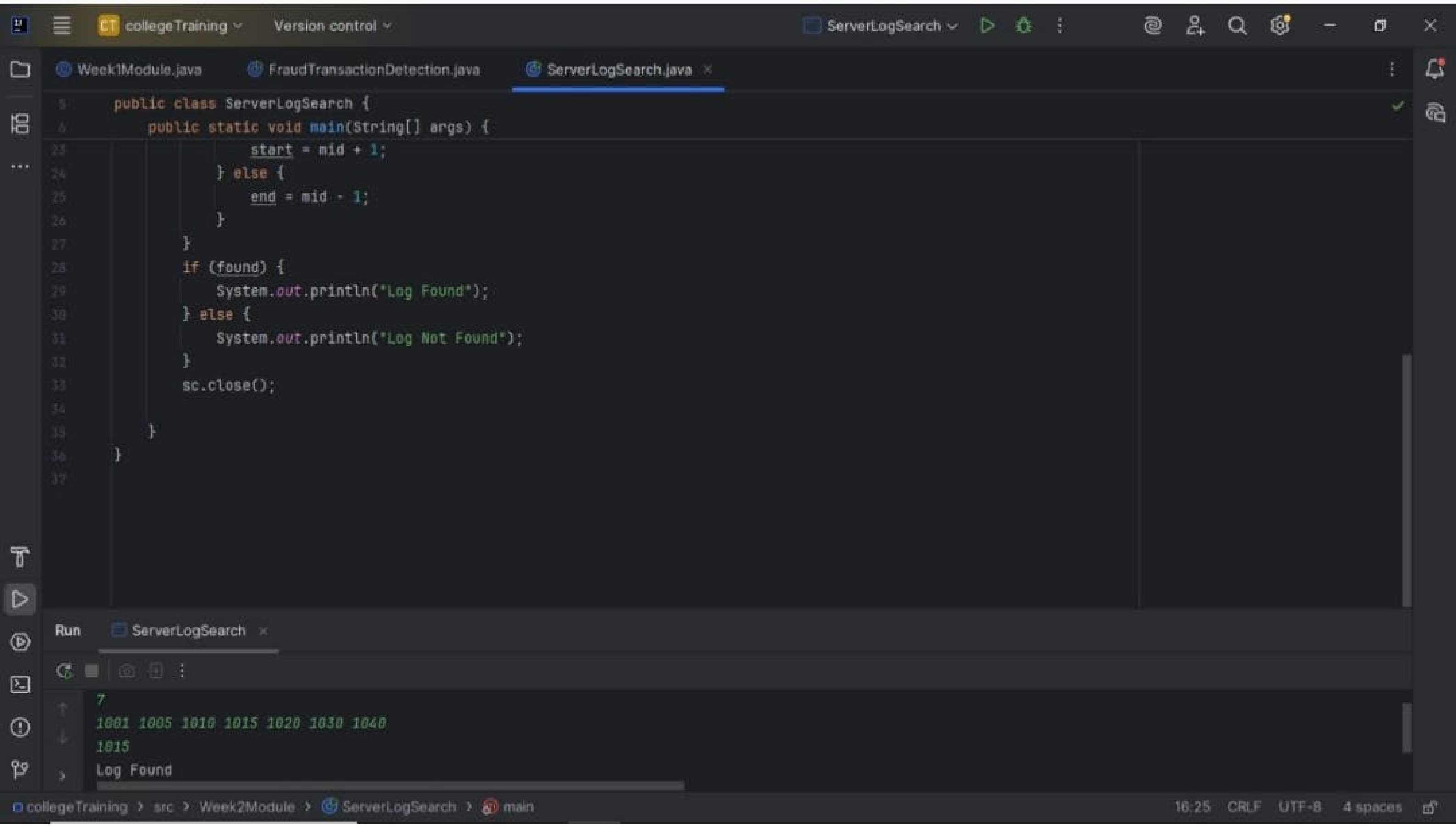
Week1Module.java FraudTransactionDetection.java ServerLogSearch.java

```java
public class ServerLogSearch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] logs = new int[n];
        for (int i = 0; i < n; i++) {
            logs[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        boolean found = false;
        int start = 0;
        int end = n - 1;
        while (start <= end) {
            int mid = start + (end - start) / 2;
            if (logs[mid] == target) {
                found = true;
                break;
            } else if (logs[mid] < target) {
                start = mid + 1;
            } else {
                end = mid - 1;
            }
```

Run ServerLogSearch

```
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found
```

collegeTraining > src > Week2Module > ServerLogSearch > main 16:25 CRLF UTF-8 4 spaces


collegeTraining
Version control
ServerLogSearch
Week1Module.java
FraudTransactionDetection.java
ServerLogSearch.java
public class ServerLogSearch {
public static void main(String[] args) {
start = mid + 1;
} else {
end = mid - 1;
}
}
if (found) {
System.out.println("Log Found");
} else {
System.out.println("Log Not Found");
}
sc.close();
}
}
Run
ServerLogSearch
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found
collegeTraining › src › Week2Module › ServerLogSearch › main
16:25 CRLF UTF-8 4 spaces

3. Sort employee salaries [bubble sort]

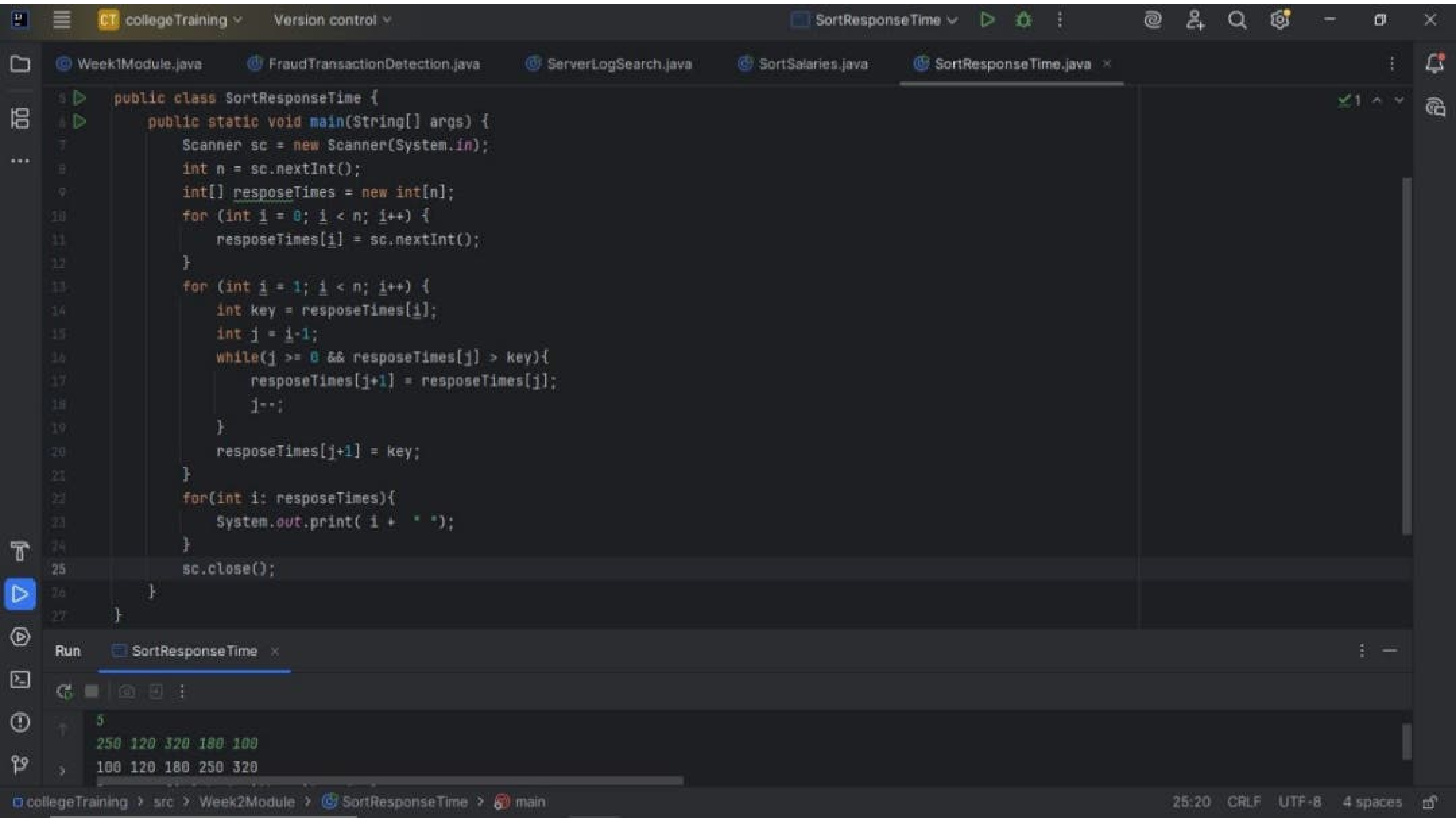
```
Public class sort salaries {
  Public static void main (string [ ] args) {
    Scanner Sc= new scanner (system.in);
    int n= sc.nextInt ();
    int [ ] salaries = new int [n];
    for (int i= 0; i< n; i++) {
      salaries [i] = sc.nextInt ();

    }
    for (int i= 0; i<n -1 ; i++) {
      for (int j= 0; j<n-i-1; j++) {
        if ( salaries [j] > salaries [j+1]) {
          int temp = salaries [j];
          salaries [j] = salaries [j+1];
          salaries [j+1] = temp;

        }
      }
    }
    for (int i : salaries) {
      system.out.print (i+"  ");

    }
    sc.close ();
  }

}
```

```java
public class SortSalaries {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] salaries = new int[n];
        for (int i = 0; i < n; i++) {
            salaries[i] = sc.nextInt();
        }
        for (int i = 0; i < n-1; i++) {
            for (int j = 0; j < n-i-1; j++) {
                if(salaries[j] > salaries[j+1]){
                    int temp = salaries[j];
                    salaries[j] = salaries[j+1];
                    salaries[j+1] =temp;
                }
            }
        }
        for(int i : salaries){
            System.out.print(i + " ");
        }sc.close();
    }
}
```

```
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
Process finished with exit code 0
```

4. Sort website Response Time (Insertion sort)

```
Public class sort Response Time {
    public static void main ( String [ ] args ) {
        Scanner SC = new scanner ( System . in );
        int n = sc . nextInt ( );
        int [ ] respose Times = new Int [n];
        for (int i = 0; i<n; i++) {
            resposeTimes [i] = sc. nextInt ( );
        }
        for (int i = 1; i<n; i++) {
            int key = responseTimes [i];
            int j = i-1;
            while (j>= 0 && respose Times [j] > key ) {
                respose Times [j+1] = respose Times [j];
                j--;
            }
            resposeTimes [j+1] = key;
        }
        for (int i : resposeTimes) {
            system . out . print (i+ ." ");
        }
        Sc. close ();
    }
}
```


collegeTraining
Version control
SortResponseTime
Week1Module.java
FraudTransactionDetection.java
ServerLogSearch.java
SortSalaries.java
SortResponseTime.java
public class SortResponseTime {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] resposeTimes = new int[n];
        for (int i = 0; i < n; i++) {
            resposeTimes[i] = sc.nextInt();
        }
        for (int i = 1; i < n; i++) {
            int key = resposeTimes[i];
            int j = i-1;
            while(j >= 0 && resposeTimes[j] > key){
                resposeTimes[j+1] = resposeTimes[j];
                j--;
            }
            resposeTimes[j+1] = key;
        }
        for(int i: resposeTimes){
            System.out.print(i + " ");
        }
        sc.close();
    }
}
Run
SortResponseTime
5
250 120 320 180 100
100 120 180 250 320
collegeTraining › src › Week2Module › SortResponseTime › main
25:20 CRLF UTF-8 4 spaces

5. Prioritize Bug Reports (selection sort)

```
Public class BugReports {
  Public static void main (string [] args) {
    Scanner sc = new scanner (system.in);
    int [] priorityscore = new int [n];
    for (int i = 0; i<n; i++) {
      priority score [i] = sc.nextInt ();
    }
    for (int i = 0; i<n-1; i++) {
      int min = i;
      for (int j = i+1; j<n; j++) {
        if (priorityscore [j] < priorityscore[min]) {
          min = j;
        }
      }
      int temp = priority score [i];
      priorityscore [i] = priorityscore [min];
      priority score [min] = temp;
    }
    for (int i : priority score) {
      system.out.print (i + " ");
    }
    sc.close ();
  }
}
```

collegeTraining Version control BugReports

Week1Module.java | FraudTransactionDetection.java | ServerLogSearch.java | SortSalaries.java | SortResponseTime.java | BugReports.java

```java
public class BugReports {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] priorityScore = new int[n];
        for (int i = 0; i < n; i++) {
            priorityScore[i] = sc.nextInt();
        }
        for (int i = 0; i < n-1; i++) {
            int min = i;
            for (int j = i+1; j < n; j++) {
                if(priorityScore[j] < priorityScore[min]){
                    min = j;
                }
            }
            int temp = priorityScore[i];
            priorityScore[i] = priorityScore[min];
            priorityScore[min] = temp;
        }
        for(int i : priorityScore){
            System.out.print(i + " ");
        }sc.close();
    }
```

Run BugReports

```
5
9 2 8 1 5
1 2 5 8 9
```

collegeTraining › src › Week2Module › BugReports › main   16:59 CRLF UTF-8 4 spaces